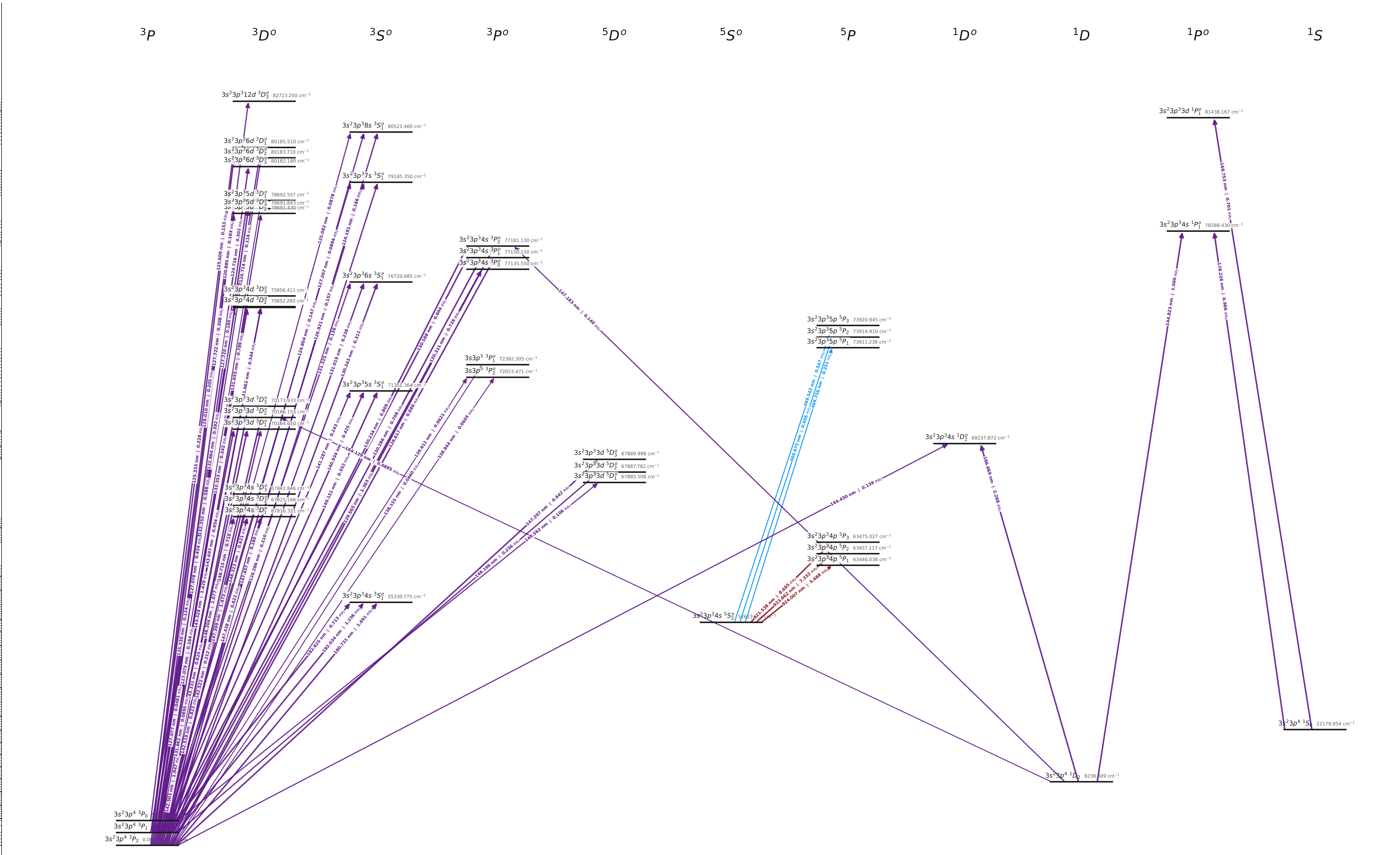


Sulfur

S (S I). Strongest classified transitions below 2 μm (NIST): 44 levels, 70 lines.

S⁺ ionisation limit 83559.17 cm⁻¹ = 10.36002 eV (119.676 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ P ₂ ^o	129.5653	-	2313.8321	1.363	3.460e+08	-	-	-
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ P ₁ ^o	129.6174	-	2312.9033	0.686	1.460e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4s ³ P ₂ ^o	130.2336	-	2301.9586	0.809	1.200e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4s ³ P ₁ ^o	130.2862	-	2301.0298	0.708	1.530e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4s ³ P ₀ ^o	130.3110	-	2300.5921	0.726	4.830e+08	-	-	-
3s ² 3p ⁴ ³ P ₀ – 3s ² 3p ³ 4s ³ P ₁ ^o	130.5883	-	2295.7073	0.804	1.960e+08	-	-	-
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 3d ³ D ₃ ^o	142.5031	-	2103.7616	1.947	3.790e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 3d ³ D ₃ ^o	143.3279	-	2091.6549	1.429	2.810e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 3d ³ D ₁ ^o	143.3310	-	2091.6086	0.825	1.560e+08	-	-	-
3s ² 3p ⁴ ³ P ₀ – 3s ² 3p ³ 3d ³ D ₁ ^o	143.6967	-	2086.2861	0.954	2.070e+08	-	-	-
3s ² 3p ⁴ ¹ D ₂ – 3s ² 3p ³ 4s ¹ P ₁ ^o	144.8230	-	2070.0616	1.500	5.000e+08	-	-	-
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ D ₃ ^o	147.3995	-	2033.8773	1.473	1.960e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4s ³ D ₃ ^o	148.3039	-	2021.4739	1.077	1.440e+08	-	-	-
3s ² 3p ⁴ ¹ D ₂ – 3s ² 3p ³ 4s ¹ D ₂ ^o	166.6687	-	1798.7326	2.288	4.580e+08	-	-	-
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ S ₁ ^o	180.7312	-	1658.7749	1.691	3.270e+08	-	-	-
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4s ³ S ₁ ^o	182.0342	-	1646.9014	1.236	1.710e+08	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).